

Multiple Choice

1. **Answer:** A

Options: A) They have flowers that bloom for several years before producing seeds; B) They have a unique leaf structure; C) They can survive in extremely cold climates; D) They produce large seeds; E) Their seeds can remain dormant for up to a century before germinating

Note: Correct option: A - They have flowers that bloom for several years before producing seeds

2. **Answer:** A

Options: A) The lateral-line system in fishes helps with the digestion of food.; B) The lateral-line system in fishes is used for excreting waste materials.; C) The lateral-line system in fishes is involved in the secretion of hormones.; D) The lateral-line system in fishes is used for breathing.; E) The lateral-line system in fishes is used for vision.

Note: Correct option: A - The lateral-line system in fishes helps with the digestion of food.

3. **Answer:** B

Options: A) dog, robin, fish; B) robin, dog; C) lizard, frog; D) fish, robin; E) fish, frog

Note: Correct option: B - robin, dog

4. **Answer:** A

Options: A) Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.; B) Earthworms and humans use lungs, frogs and fish use gills.; C) All use gills for obtaining oxygen.; D) Earthworms and humans use gills, frogs use skin, and fish use lungs.; E) Earthworms use lungs, frogs use gills, fish use skin, and humans use gills.

Note: Correct option: A - Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.

5. **Answer:** A

Options: A) By assessing the fossil content found within the rock layers; B) By comparing the density of the rock to a known age database; C) By examining the color and texture of the rock; D) By counting the layers of the rock; E) By measuring the amount of carbon-14 remaining in the rock

Note: Correct option: A - By assessing the fossil content found within the rock layers

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'insulin decreases storage of glycogen'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'parallel selection'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Adenosine triphosphate'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Destruction of the ozone layer'.

Long Form Response

11. **Answer:** A few drug-resistant viruses were present at the start of treatment, and natural selection increased their frequency.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Presence of vessels in both xylem and phloem, autospory, reduced and independent gametophyte, embryo without a suspensor. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key